Ex 5.2.5

Prove Corollary 5.18:
Suppose that D is a domain with $a \in D$. Suppose also that $f \in \mathcal{H}(D \setminus \{a\})$. Show that f has a pole of order m at a if and only if there is a $g \in \mathcal{H}(D)$ such that $g(a) \neq 0$ and

$$f(z) = \frac{g(z)}{(z-a)^m} \quad \text{for all } z \in D \setminus \{a\}.$$

Pf. **(⇒)** If f has a pole of order m at a. By Definition 5.14, there exist constants $b_1, \ldots, b_m \in \mathbb{C}$ with $b_m \neq 0$ s.t.

$$r(z) = f(z) - \sum_{j=1}^{m} \frac{b_j}{(z-a)^j}$$

has a removable singularity at a. We can improve r to be holomorphic at a also, then $r \in \mathcal{H}(D)$. For $z \in D \setminus \{a\}$, multiplying by $(z-a)^m$, we have

$$(z-a)^m f(z) = (z-a)^m r(z) + \sum_{j=1}^{m} b_j (z-a)^{m-j}.$$

Let $g : D \to \mathbb{C}$ be

$$g(z) := (z-a)^m r(z) + b_m + b_{m-1}(z-a) + \cdots + b_1 (z-a)^{m-1}.$$

Each sum is holomorphic on D, so $g \in \mathcal{H}(D)$. Also $g(a) = b_m \neq 0$. On $D \setminus \{a\}$ we have $g(z) = (z-a)^m f(z)$, so $f(z) = g(z)/(z-a)^m$.

(⇐) If $g \in \mathcal{H}(D)$, $g(a) \neq 0$, and $f(z) = g(z)/(z-a)^m$ on $D \setminus \{a\}$. By Theorem 4.12 (Holomorphic Functions are Analytic), g has a Taylor expansion convergent in some $D_r(a) \subset D$:

$$g(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$
$$c_0 = g(a) \neq 0.$$

For $z \in D_r(a) \setminus \{a\}$ we then have

$$f(z) = \frac{g(z)}{(z-a)^m} = \sum_{n=0}^{\infty} c_n (z-a)^{n-m} = \sum_{j=1}^{m} \frac{c_{m-j}}{(z-a)^j} + \sum_{n=m}^{\infty} c_n (z-a)^{n-m}.$$

Let $b_j := c_{m-j}$ for $j \in \{1, \ldots, m\}$. Then $b_m = c_0 = g(a) \neq 0$. Thus,

$$r(z) := f(z) - \sum_{j=1}^{m} \frac{b_j}{(z-a)^j} = \sum_{k=0}^{\infty} c_{k+m} (z-a)^k.$$

The RHS is the value of an analytic function on $D_r(a)$, so r has a removable singularity at a. By Definition 5.14, f has a pole of order m at a.

Hence shown that f has a pole of order m at a iff $\exists g \in \mathcal{H}(D)$, $g(a) \neq 0$, and $f(z) = g(z)/(z-a)^m$. □

Ex 5.2.11

Suppose that f has an isolated singularity at a.

(a) Suppose that f has an essential singularity at a. Show that $g(z) = \exp(f(z))$ has an essential singularity at a.

(b) Suppose that f has a pole at a. Show that $g(z) = \exp(f(z))$ has an essential singularity at a. (Hint: consider Ex 5.2.9.)

(c) Suppose that $g(z) = \exp(f(z))$ has a removable singularity at a. Show that f has a removable singularity at a.

Pf. Since exp is entire, $g(z) = \exp(f(z))$ is holomorphic on the same modified neighborhood of a as f. Therefore, g has an isolated singularity at a, of exactly one type by Def 5.20: removable, pole, or essential.
We also know that $|g(z)| = e^{\operatorname{Re} f(z)}$.

(c) If g has a removable singularity at a. By Proposition 5.13(d), there exists $s, M > 0$ such that $|g(z)| \leqslant M$ for $z \in D'_s(a)$. Then $\operatorname{Re} f(z) \leqslant \ln M$ on $D'_s(a)$. We can consider the other two singularity types for f:

- If f has a pole at a, then by Ex 5.2.9, $\sup_{z\in D'_s(a)} \operatorname{Re} f(z) = \infty$, which contradicts $\operatorname{Re} f \leqslant \ln M$.
- If f has an essential singularity at a, then by Theorem 5.21 (Casorati–Weierstrass), $f(D'_s(a))$ is dense in $\mathbb{C}$, so $\operatorname{Re} f$ takes arbitrarily large positive values, which again contardicts $\operatorname{Re} f \leqslant \ln M$.

Therefore f has to be a removable singularity at a. □

(a) If f has an essential singularity at a. We can consider the other two singularity types for g:

- If g has a removable singularity at a, then part (c) implies that f has a removable singularity at a, which contradicts our assumption.
- If g has a pole at a, then by Proposition 5.19, $|g(z)| \to \infty$, i.e. $\operatorname{Re} f(z) \to \infty$. So $\forall M > 0$, $\exists \delta > 0$ s.t. $\operatorname{Re} f(z) > M$ on $D'_\delta(a)$. But then $f(D'_\delta(a)) \subset \{w : \operatorname{Re} w > M\}$, and the complement contains an open disk, i.e. not dense. This contradicts Theorem 5.21 (C-W) on f.

Therefore g has to be an essential singularity at a. □

(b) If f has a pole at a. We can consider the other two singularity types for g:

- If g has a removable singularity at a, then part (c) gives f is removable, which is a contradiction.
- If g has a pole at a, then $|g(z)| \to \infty$, so $\operatorname{Re} f(z) \to \infty$. In particular $\operatorname{Re} f$ is bounded below on a punctured neighborhood of a. But Ex 5.2.9 gives $\sup_{z\in D'_s(a)} \big(-\operatorname{Re} f(z)\big) = \infty$, i.e. $\operatorname{Re} f$ is unbounded below on every modified neighborhood of a, which is a contradiction again.

Therefore g has to be an essential singularity at a. □

Ex 5.2.12

Suppose that $f \in \mathcal{H}(D)$. Let $\{a_1, \ldots, a_n\} \subset D$ be zeros of f such that a_k has multiplicity m_k. By considering the singularities of $g(z) = f(z)(z-a_1)^{-m_1} \cdots (z-a_n)^{-m_n}$, show that

$$f(z) = (z-a_1)^{m_1} \cdots (z-a_n)^{m_n} g(z) \quad \text{for all } z \in D$$

with $g \in \mathcal{H}(D)$ and such that $g(a_j) \neq 0$ for $1 \leqslant j \leqslant n$. If $Z(f) = \{a_1, \ldots, a_n\}$ are all the zeros of f in D, conclude that g does not vanish on D as asserted in part (b) of Ex 5.1.5.

Pf. Let $g : D \to \mathbb{C}$ be

$$g(z) := \frac{f(z)}{(z-a_1)^{m_1} \cdots (z-a_n)^{m_n}}, \quad z \in D \setminus \{a_1, \ldots, a_n\}.$$

By Prop 2.2, this is holomorphic on $D \setminus \{a_1, \ldots, a_n\}$ as the quotient of holomorphic functions with denominator that does not approach zero. So g has isolated singularities for all a_k.
For a given k, by Lemm 5.4, since f has a zero of order m_k at a_k, $\exists h_k \in \mathcal{H}(D)$ with $h_k(a_k) \neq 0$ s.t.

$$f(z) = (z-a_k)^{m_k} h_k(z)$$

for $z \in D$. We can substitute this into g to get,

$$g(z) = \frac{(z-a_k)^{m_k} h_k(z)}{(z-a_k)^{m_k} \prod_{j \neq k} (z-a_j)^{m_j}} = \frac{h_k(z)}{\prod_{j \neq k} (z-a_j)^{m_j}}$$

for $z \in D \setminus \{a_1, \ldots, a_n\}$. The RHS is holomorphic at a_k (the points a_j with $j \neq k$ are distinct from a_k, so the denominator does not zero out at a_k), and the value at a_k is

$$\frac{h_k(a_k)}{\prod_{j \neq k} (a_k - a_j)^{m_j}} \neq 0,$$

since $h_k(a_k) \neq 0$ and $a_k \neq a_j$ for $j \neq k$. Hence a_k is a removable singularity of g, and the extension (still g) is holomorphic at a_k with $g(a_k) \neq 0$.
Since this holds for every k, g extends to an element of $\mathcal{H}(D)$ with $g(a_j) \neq 0$ for all j. We have,

$$f(z) = (z-a_1)^{m_1} \cdots (z-a_n)^{m_n} g(z)$$

holds on $D \setminus \{a_1, \ldots, a_n\}$, and both sides are holomorphic on D, so they equal on a set with limit points in D. By Corollary 5.7 they equal on all of D.

Lert $Z(f) = \{a_1, \ldots, a_n\}$ be the zeros of f in D. If $g(z_0) = 0$ for some $z_0 \in D$, then $z_0 \notin \{a_1, \ldots, a_n\}$ since $g(a_j) \neq 0$ for every j. Then,

$$f(z_0) = (z_0 - a_1)^{m_1} \cdots (z_0 - a_n)^{m_n} g(z_0) = 0,$$

so $z_0 \in Z(f) = \{a_1, \ldots, a_n\}$, which is a contradiction. Hence g does not zero out (i.e. vanish) on D. □

Ex 5.3.6

Suppose that f is entire. Show that f is a polynomial of at most degree n ($n \geqslant 1$) if and only if there are non-negative constants A and B such that

$$|f(z)| \leqslant A|z|^n + B \quad \text{for all } z \in \mathbb{C}.$$

Pf. **(⇒)** If $f(z) = c_0 + c_1 z + \cdots + c_n z^n$. $\forall z \in \mathbb{C}$ and $0 \leqslant k \leqslant n$, we have $|z|^k \leqslant 1 + |z|^n$ (which has the parts split as $|z| \leqslant 1$ which gives $|z|^k \leqslant 1$, and $|z| \geqslant 1$ which gives $|z|^k \leqslant |z|^n$). Therefore,

$$|f(z)| \leqslant \sum_{k=0}^{n} |c_k|\,|z|^k \leqslant \left(\sum_{k=0}^{n} |c_k|\right)(1 + |z|^n).$$

Take $A = B = \sum_{k=0}^{n} |c_k| \geqslant 0$ which gives $|f(z)| \leqslant A|z|^n + B$.

(⇐) If $|f(z)| \leqslant A|z|^n + B$ for all $z \in \mathbb{C}$. Since f is entire, by Theorem 4.12 (Holomorphic Functions are Analytic) it has a Maclaurin series,

$$f(z) = \sum_{k=0}^{\infty} a_k z^k$$
$$a_k = \frac{f^{(k)}(0)}{k!}$$

which converges on $\mathbb{C}$.

$\forall R > 0$, from the bound we have $|f(z)| \leqslant AR^n + B$ on $\overline{D_R(0)}$. By Prop 5.25 (Cauchy's Estimates), we have,

$$|a_k| = \frac{|f^{(k)}(0)|}{k!} \leqslant \frac{AR^n + B}{R^k} = AR^{n-k} + BR^{-k}$$

Also, for $k > n$, both AR^{n-k} and BR^{-k} tend to 0 as R tends to ∞, so $a_k = 0\ \forall k > n$. Therefore,

$$f(z) = a_0 + a_1 z + \cdots + a_n z^n$$

is a polynomial of degree at most n.

Hence shown that f has degree at most n iff we have $|f(z)| \leqslant A|z|^n + B, \forall z \in \mathbb{C}$ where A and B are nonnegative constants. □

Ex 5.3.9

Suppose that f and g are entire and such that $|f(z)| \leqslant |g(z)|$ for all $z \in \mathbb{C}$. What can be said about f and g? (Hint: consider $h = f/g$.) In particular, what can you conclude if $|f(z)| \leqslant A|z|^n$ for all z? (Compare with Ex 5.3.6.)

Pf. We claim that $f = cg$ for some constant $c \in \mathbb{C}$ with $|c| \leqslant 1$.

In the trivial case, if $g \equiv 0$ then $|f| \leqslant 0$ forces $f \equiv 0$, and the claim holds with $c = 0$.

Otherwise, g is nonconstant entire, and by Thm 5.6 its zero set $Z(g)$ is isolated in $\mathbb{C}$. Let

$$h(z) := \frac{f(z)}{g(z)}$$

for $z \in \mathbb{C} \setminus Z(g)$. Then by Prop 2.2, $h \in \mathcal{H}(\mathbb{C} \setminus Z(g))$ and $|h(z)| \leqslant 1$. $\forall a \in Z(g)$ we have an isolated singularity of h at which h is bounded, so by Prop 5.13(d) it is removable. By extending h for every $a \in Z(g)$, we have an entire function which is restricted by 1 on $\mathbb{C}$.

Then by Thm 5.26 (Liouville), $h \equiv c$ for some constant with $|c| \leqslant 1$. Therefore, $f(z) = cg(z)$, $\forall z \in \mathbb{C}$.

For the case where $|f(z)| \leqslant A|z|^n$, we can still use the claim to f and $g(z) = Az^n$ s.t. $|f(z)| \leqslant A|z|^n = |g(z)|$, so $f = cAz^n$ for some $|c| \leqslant 1$. Let $\alpha := cA$, then we get $f(z) = \alpha z^n$, with $|\alpha| \leqslant A$.

This is a stronger statement than Ex 5.3.6, which only gives that f is a polynomial of degree at most n. Here f must be the single monomial αz^n.

Ex 5.3.12

Suppose that f is entire and that the range of f is disjoint from the ray $(-\infty, 0]$. Show that f is constant. (Hint: let s be the branch of $z^{1/2}$ from Ex 5.3.11, and consider $g = s \circ f$.)

Pf. Using the solution from Ex 5.3.11, the function $s(z) = \exp\left(\frac{1}{2}\mathrm{Log}(z)\right)$ is an analytic branch of $z^{1/2}$ on $D^* = \mathbb{C} \setminus (-\infty, 0]$, and its range lies in $H = \{w \in \mathbb{C} : \mathrm{Re}(w) > 0\}$.

Since the range of f is disjoint from $(-\infty, 0]$, we have $f(\mathbb{C}) \subset D^*$.
Let $g := s \circ f$. By Prop 2.2 (Chain Rule), g is entire, and its range satisfies $g(\mathbb{C}) \subset s(D^*) \subset H$, so $\mathrm{Re}\, g(z) > 0$, $\forall z \in \mathbb{C}$.

Now consider $\exp(-g)$, which is a composition of entire functions and by Prop 2.2 is also entire. Then $\forall z \in \mathbb{C}$ we have,

$$|\exp(-g(z))| = e^{-\mathrm{Re}\, g(z)} < e^0 = 1.$$

So $\exp(-g)$ is bounded entire, therefore constant by Thm 5.26 (Liouville).

Therefore we have g is also constant. We can write $\exp(-g(z)) \equiv c$, the function g takes values in the discrete set $\{w \in \mathbb{C} : e^{-w} = c\}$, i.e. $2\pi i\mathbb{Z}$.
Since g is continuous and $\mathbb{C}$ is connected, g must be constant.

We have s is a branch of $z^{1/2}$, so $s(w)^2 = w$ on D^*. For all values of z we have,

$$f(z) = s(f(z))^2 = g(z)^2 = \text{constant}.$$

Therefore, f is constant. □

Ex 5.4.1

Let D be a domain. Show that (f_n) converges to f uniformly on compact subsets of D if and only if (f_n) converges uniformly to f on every closed disk $\overline{D_r(a)}$ contained in D.

Pf. **(⇒)** As in the proof for Thm 5.32, we have a closed disk $\overline{D_r(a)} \subset D$ is a compact subset of D. So uniform convergence on compact subsets of D immediately gives uniform convergence on $\overline{D_r(a)}$.

(⇐) If f_n converges to f uniformly on every closed disk $\overline{D_r(a)} \subset D$. Let $K \subset D$ be compact. For each $a \in K$ (because D is open and $a \in D$), we have $r_a > 0$ s.t. $\overline{D_{r_a}(a)} \subset D$.

The open disks $\{D_{r_a}(a) : a \in K\}$ form a cover of K which is also open. By compactness of K, finitely many cover K. I.e. $\exists a_1, \ldots, a_p \in K$ s.t.

$$K \subseteq \bigcup_{i=1}^{p} D_{r_{a_i}}(a_i) \subseteq \bigcup_{i=1}^{p} \overline{D_{r_{a_i}}(a_i)} \subset D.$$

Let $\varepsilon > 0$. Take that for each i, $\exists N_i \in \mathbb{N}$ s.t. $n \geqslant N_i$, which implies $|f_n(z) - f(z)| < \varepsilon$ for all $z \in \overline{D_{r_{a_i}}(a_i)}$.
Let $N := \max(N_1, \ldots, N_p)$. For $n \geqslant N$ and $\forall z \in K$, z lies in some $\overline{D_{r_{a_i}}(a_i)}$, which implies $|f_n(z) - f(z)| < \varepsilon$.
Therefore, we have f_n converges to f uniformly on K.

Hence shown that (f_n) converges to f uniformly on compact subsets of D iff (f_n) converges uniformly to f on every closed disk $\overline{D_r(a)}$ contained in D. □

Ex 5.4.2

Show that the Taylor series for $f \in \mathcal{H}(D_R(a))$ about a converges uniformly to f on compact subsets of $D_R(a)$.

Pf. From Thm 4.12 (Holomorphic Functions are Analytic), f has a Taylor expansion,

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$
$$c_n = \frac{f^{(n)}(a)}{n!},$$

which converges pointwise on $D_R(a)$.

Let $K \subset D_R(a)$ be compact. The continuous function $z \mapsto |z-a|$ is maximum on K, and the maximum is strictly less than R (since $K \subset D_R(a)$).
Take ρ and s with $\max_{z\in K} |z-a| \leqslant \rho < s < R$, s.t. $K \subseteq \overline{D_\rho(a)} \subset D_s(a) \subset D_R(a)$.
Since the Taylor series converges at every point of $D_s(a)$ (as it converges on the larger $D_R(a)$), it converges at any z with $|z-a| = s$, so $|c_n|s^n \to 0$. Therefore, $\exists M > 0$ with $|c_n|s^n \leqslant M$, $\forall n$.

For $z \in \overline{D_\rho(a)}$, we have

$$|c_n(z-a)^n| \leqslant |c_n|\rho^n = (|c_n|s^n)\left(\frac{\rho}{s}\right)^n \leqslant M\left(\frac{\rho}{s}\right)^n$$

Since $0 < \rho/s < 1$, $\sum_{n=0}^{\infty} M(\rho/s)^n$ is a geometric series which is convergent. By the Weierstrass M-test, $\sum c_n(z-a)^n$ converges uniformly on $\overline{D_\rho(a)}$, and hence on K.

Since K was an arbitrary compact subset of $D_R(a)$, we have that the Taylor series converges uniformly to f on compact subsets of $D_R(a)$. □

Ex 5.4.5

Let (f_n) be a sequence of holomorphic functions on an open set D converging pointwise to a function f on D. Suppose that each point $a \in D$ has a neighborhood $D_r(a)$ so that (f_n) is uniformly bounded on $D_r(a)$. Show that (f_n) converges to f uniformly on compact subsets of D and hence that f is holomorphic on D. (Hint: show that (f_n) is uniformly Cauchy. You may need the Dominated Convergence Theorem in the form of Ex 5.4.4.)

Pf. First, show that every $a \in D$ has a closed disk $\overline{D_{\rho/2}(a)} \subset D$ on which (f_n) is uniformly Cauchy.

Take a fixed $a \in D$. Assume there are $r > 0$ and $M > 0$ s.t. $D_r(a) \subset D$ and $|f_n(z)| \leqslant M$, $\forall z \in D_r(a)$ and $\forall n \in \mathbb{N}$. Take any $0 < \rho < r$ and let $\gamma(t) = a + \rho e^{it}$, $t \in [0, 2\pi]$.
Since $D_r(a)$ is convex, by Lemm 4.4 we have $\gamma^* \subset D_r(a)$, $\mathrm{Ind}_\gamma(z) = 1$ for $z \in D_\rho(a)$. With Thm 4.10 (Cauchy Integral Formula) and $f_n \in \mathcal{H}(D_r(a))$, we have $\forall z \in D_\rho(a)$,

$$f_n(z) = \frac{1}{2\pi i} \int_\gamma \frac{f_n(w)}{w - z}\, dw.$$

Now restrict to $K := \overline{D_{\rho/2}(a)}$. For $z \in K$ and $w \in \gamma^*$, $|w - z| \geqslant \rho - |z - a| \geqslant \rho/2$. Hence for any $m, n \in \mathbb{N}$ and $z \in K$,

$$|f_n(z) - f_m(z)| = \left| \frac{1}{2\pi i} \int_\gamma \frac{f_n(w) - f_m(w)}{w - z}\, dw \right| \leqslant \frac{1}{2\pi} \frac{2}{\rho} \int_\gamma |f_n(w) - f_m(w)|\, |dw|.$$

I.e., RHS is independent of $z \in K$.

On γ^*, $|f_n - f| \leqslant |f_n| + |f| \leqslant 2M$ (the bound $|f| \leqslant M$ follows from $|f_n| \leqslant M$ and pointwise convergence), and $f_n - f$ tends to 0 pointwise on γ^*. From Ex 5.4.4 (Dominated Convergence Theorem),

$$\int_\gamma |f_n(w) - f(w)|\, |dw| \to 0 \quad \text{as } n \to \infty.$$

Therefore using the triangle inequality,

$$\int_\gamma |f_n - f_m|\, |dw| \leqslant \int_\gamma |f_n - f|\, |dw| + \int_\gamma |f - f_m|\, |dw| \to 0$$

as m, n tend to ∞. So (f_n) is uniformly Cauchy on $K = \overline{D_{\rho/2}(a)}$.

Now, uniform Cauchy on K implies uniform convergence on K to some continuous limit. We use our assumption of pointwise convergence to take the limit as f, so f_n tends to f uniformly on K.
We have shown $\forall a \in D$ has an open disk $D_{\rho_a/2}(a)$ on whose closure f_n tends to f uniformly.

Let $L \subset D$ be compact. Then the disks $\{D_{\rho_a/2}(a) : a \in L\}$ also form a cover of L which is open. Then by compactness, we have that finitely many cover L, i.e.

$$L \subseteq \bigcup_i \overline{D_{\rho_{a_i}/2}(a_i)}$$

Uniform convergence on each piece gives uniform convergence on L as a whole.

Since f_n tends to f uniformly on compact subsets of D and each $f_n \in \mathcal{H}(D)$, by Thm 5.32, we have $f \in \mathcal{H}(D)$. $\square$